

CC LAB-10

NAME: S SHREYAS

SRN : PES1UG22CS494

TASK-1

```
(base) shrey@shrey-VirtualBox:~/CC_E4_PES1UG22CS494$ pyenv versions
system
* 3.6.8 (set by /home/shrey/.python-version)
(base) shrey@shrey-VirtualBox:~/CC_E4_PES1UG22CS494$ pyenv local 3.6.8
(base) shrey@shrey-VirtualBox:~/CC_E4_PES1UG22CS494$ python --version
Python 3.6.8
(base) shrey@shrey-VirtualBox:~/CC_E4_PES1UG22CS494$ ls -a
.  ..  .python-version  raftos
(base) shrev@shrev-VirtualBox:~/CC_E4_PES1UG22CS494$
```

(base) root@shrey-VirtualBox:/home/shrey/CC_E4_PES1UG22CS494/CC_Lab9_Raft#	(base) root@shrey-VirtualBox:/home/shrey/CC_E4_PES1UG22CS494/CC_Lab9_Raft#
(base) root@shrey-VirtualBox:/home/shrey/CC_E4_PES1UG22CS494/CC_Lab9_Raft# tmux split-window S -v unknown command: split-windows (base) root@shrey-VirtualBox:/home/shrey/CC_E4_PES1UG22CS494/CC_Lab9_Raft# tmux split-window -v (base) root@shrey-VirtualBox:/home/shrey/CC_E4_PES1UG22CS494/CC_Lab9_Raft#	(base) root@shrey-VirtualBox:/home/shrey/CC_E4_PES1UG22CS494/CC_Lab9_Raft# tmux split-window w -v (base) root@shrey-VirtualBox:/home/shrey/CC_E4_PES1UG22CS494/CC_Lab9_Raft#
(base) root@shrey-VirtualBox:/home/shrey/CC_E4_PES1UG22CS494/CC_Lab9_Raft#	(base) root@shrey-VirtualBox:/home/shrey/CC_E4_PES1UG22CS494/CC_Lab9_Raft#

[PES2UG22CS494@bash*] shrev-VirtualBox 11:25 10-Apr-23

TASK -2

<pre>(base) shrey@shrey-VirtualBox:~/CC_E4_PES1UG22CS494/raftos\$ python node.py --node 1</pre>	<pre>(base) shrey@shrey-VirtualBox:~/CC_E4_PES1UG22CS494/raftos\$ tail -f node1_CUSTOMLOG.log Leader is now: 127.0.0.1:8000 Done writing VARIABLE Check the .log and .state_machine files!</pre>
<pre>(base) shrey@shrey-VirtualBox:~/CC_E4_PES1UG22CS494/raftos\$ python node.py --node 2</pre>	<pre>(base) shrey@shrey-VirtualBox:~/CC_E4_PES1UG22CS494/raftos\$ tail -f node2_CUSTOMLOG.log Leader is now: 127.0.0.1:8000 Leader is now: 127.0.0.1:8002 Leader is now: 127.0.0.1:8000 Leader is now: 127.0.0.1:8002</pre>
<pre>(base) shrey@shrey-VirtualBox:~/CC_E4_PES1UG22CS494/raftos\$ python node.py --node 3</pre>	<pre>(base) shrey@shrey-VirtualBox:~/CC_E4_PES1UG22CS494/raftos\$ tail -f node3_CUSTOMLOG.log Leader is now: 127.0.0.1:8000 Leader is now: 127.0.0.1:8002</pre>

Screenshot 2a

<pre>Open 127.0.0.1_8001.log {"term": 4, "command": {"VARIABLE": "PES1UG22CS494"}, "hashcode": "*****"} {"term": 10, "command": {"VARIABLE": "HashCode"}, "hashcode": "*****"} {"term": 34, "command": {"VARIABLE": "HashCode"}, "hashcode": "*****"}</pre>	<pre>Open 127.0.0.1_8001.log {"term": 4, "command": {"VARIABLE": "PES1UG22CS494"}, "hashcode": "*****"} {"term": 10, "command": {"VARIABLE": "HashCode"}, "hashcode": "*****"} {"term": 34, "command": {"VARIABLE": "HashCode"}, "hashcode": "*****"}</pre>	<pre>Open 127.0.0.1_8002.log {"term": 4, "command": {"VARIABLE": "PES1UG22CS494"}, "hashcode": "*****"} {"term": 10, "command": {"VARIABLE": "HashCode"}, "hashcode": "*****"} {"term": 34, "command": {"VARIABLE": "HashCode"}, "hashcode": "*****"}</pre>
<pre>Open 127.0.0.1_8000.state_machine {"VARIABLE": "HashCode"}, "hashcode": "*****"}</pre>	<pre>Open 127.0.0.1_8001.state_machine {"VARIABLE": "HashCode"}, "hashcode": "*****"}</pre>	<pre>Open 127.0.0.1_8002.state_machine {"VARIABLE": "HashCode"}, "hashcode": "*****"}</pre>

Screenshot 2b

TASK-3

```
shrey@shrey-VirtualBox: ~/CC_E4_PES1UG22C5494/raftos
File "/home/shrey/CC_E4_PES1UG22C5494/raftos/raftos/replicator.py", line 42, in get
self.value = await State.get_value(self.name)
File "/home/shrey/CC_E4_PES1UG22C5494/raftos/raftos/state.py", line 481, in wrapped
'Leader is {}'.format(cls.leader or 'not chosen yet')
raftos.exceptions.NotALeaderException: Leader is 127.0.0.1:8002!

during handling of the above exception, another exception occurred:

Traceback (most recent call last):
  File "node.py", line 87, in <module>
    loop.run_until_complete(run(loop))
  File "/home/shrey/.pyenv/versions/3.6.8/lib/python3.6/asyncio/base_events.py", line 484, in
run_until_complete
    return future.result()
  File "node.py", line 81, in run
    await VARIABLE.set('PES1UG22C5494')
  File "/home/shrey/CC_E4_PES1UG22C5494/raftos/raftos/replicator.py", line 49, in set
    await State.set_value(self.name, value)
  File "/home/shrey/CC_E4_PES1UG22C5494/raftos/raftos/state.py", line 481, in wrapped
    'Leader is {}'.format(cls.leader or 'not chosen yet')
raftos.exceptions.NotALeaderException: Leader is 127.0.0.1:8002!
(base) shrey@shrey-VirtualBox: ~/CC_E4_PES1UG22C5494/raftos$

self._run_once()
File "/home/shrey/.pyenv/versions/3.6.8/lib/python3.6/asyncio/base_events.py", line 1415,
in _run_once
    event_list = self._selector.select(timeout)
File "/home/shrey/.pyenv/versions/3.6.8/lib/python3.6/selectors.py", line 445, in select
    fd_event_list = self._epoll.poll(timeout, max_ev)
KeyboardInterrupt
(base) shrey@shrey-VirtualBox: ~/CC_E4_PES1UG22C5494/raftos$

(base) shrey@shrey-VirtualBox: ~/CC_E4_PES1UG22C5494/raftos$ python node.py --node 3

(base) shrey@shrey-VirtualBox: ~/CC_E4_PES1UG22C5494/raftos$ tail -f node1_CUSTOMLOG.log
Leader is now: 127.0.0.1:8000
Done writing VARIABLE
Check the .log and .state_machine files!tail: node1_CUSTOMLOG.log: file truncated
Leader is now: 127.0.0.1:8001
Leader is now: 127.0.0.1:8000
Leader is now: 127.0.0.1:8000

Leader is now: None
Leader is now: 127.0.0.1:8000
Leader is now: None
Leader is now: 127.0.0.1:8001
Leader is now: 127.0.0.1:8001
Done writing VARIABLE
Check the .log and .state_machine files!tail: node2_CUSTOMLOG.log: file truncated
Leader is now: 127.0.0.1:8000
Leader is now: None
Leader is now: 127.0.0.1:8002
```

Screenshot 3a

```
Open 127.0.0.1_8001.log
{"term": 4, "command": {"VARIABLE": "PES1UG22C5494"}}
{"term": 10, "command": {"VARIABLE": "HashCode"}}
{"term": 34, "command": {"VARIABLE": "HashCode"}}

Open 127.0.0.1_8001.state_machine
{"VARIABLE": "HashCode"}

Open 127.0.0.1_8002.log
{"term": 4, "command": {"VARIABLE": "PES1UG22C5494"}}
{"term": 10, "command": {"VARIABLE": "HashCode"}}
{"term": 34, "command": {"VARIABLE": "HashCode"}}

Open 127.0.0.1_8002.state_machine
{"VARIABLE": "HashCode"}
```

Screenshot 3b

```
shrey@shrey-VirtualBox: ~/CC_E4_PES1UG22C5494/raftos
File "/home/shrey/CC_E4_PES1UG22C5494/raftos/raftos/replicator.py", line 42, in get
self.value = await State.get_value(self.name)
File "/home/shrey/CC_E4_PES1UG22C5494/raftos/raftos/state.py", line 481, in wrapped
'Leader is {}'.format(cls.leader or 'not chosen yet')
raftos.exceptions.NotLeaderException: Leader is 127.0.0.1:8002!

during handling of the above exception, another exception occurred:

Traceback (most recent call last):
  File "node.py", line 87, in <module>
    loop.run_until_complete(run(loop))
  File "/home/shrey/.pyenv/versions/3.6.8/lib/python3.6/asyncio/base_events.py", line 484, in
run_until_complete
    return future.result()
  File "node.py", line 81, in run
    await VARIABLE.set('PES1UG22C5494')
  File "/home/shrey/CC_E4_PES1UG22C5494/raftos/raftos/replicator.py", line 49, in set
    await State.set_value(self.name, value)
  File "/home/shrey/CC_E4_PES1UG22C5494/raftos/raftos/state.py", line 481, in wrapped
    'Leader is {}'.format(cls.leader or 'not chosen yet')
raftos.exceptions.NotLeaderException: Leader is 127.0.0.1:8002!
(base) shrey@shrey-VirtualBox: ~/CC_E4_PES1UG22C5494/raftos$

self._run_once()
File "/home/shrey/.pyenv/versions/3.6.8/lib/python3.6/asyncio/base_events.py", line 1415, in
_run_once
    event_list = self._selector.select(timeout)
File "/home/shrey/.pyenv/versions/3.6.8/lib/python3.6/selectors.py", line 445, in select
    fd_event_list = self._epoll.poll(timeout, max_ev)
KeyboardInterrupt
(base) shrey@shrey-VirtualBox: ~/CC_E4_PES1UG22C5494/raftos$

(base) shrey@shrey-VirtualBox: ~/CC_E4_PES1UG22C5494/raftos$ python node.py --node 3
```

```
(base) shrey@shrey-VirtualBox: ~/CC_E4_PES1UG22C5494/raftos$ tail -f node1_CUSTOMLOG.log
Leader is now: 127.0.0.1:8000
Done writing VARIABLE
Check the .log and .state_machine files!tail: node1_CUSTOMLOG.log: file truncated
Leader is now: 127.0.0.1:8001
Leader is now: 127.0.0.1:8000

Leader is now: None
Leader is now: 127.0.0.1:8000
Leader is now: None
Leader is now: 127.0.0.1:8001
Done writing VARIABLE
Check the .log and .state_machine files!tail: node2_CUSTOMLOG.log: file truncated
Leader is now: 127.0.0.1:8000
Leader is now: None
Leader is now: 127.0.0.1:8000

Leader is now: None
Leader is now: 127.0.0.1:8000
Leader is now: 127.0.0.1:8001
Leader is now: None
Leader is now: 127.0.0.1:8001
tail: node3_CUSTOMLOG.log: file truncated
Leader is now: 127.0.0.1:8000
Leader is now: None
Leader is now: 127.0.0.1:8002
```

Screenshot 4a

```
Open 127.0.0.1_8000.log
{"term": 4, "command": {"VARIABLE": "PES1UG22C5494"}}
{"term": 10, "command": {"VARIABLE": "HashCode"}}
{"term": 34, "command": {"VARIABLE": "HashCode"}}

Open 127.0.0.1_8001.log
{"term": 4, "command": {"VARIABLE": "PES1UG22C5494"}}
{"term": 10, "command": {"VARIABLE": "HashCode"}}
{"term": 34, "command": {"VARIABLE": "HashCode"}}

Open 127.0.0.1_8002.log
{"term": 4, "command": {"VARIABLE": "PES1UG22C5494"}}
{"term": 10, "command": {"VARIABLE": "HashCode"}}
{"term": 34, "command": {"VARIABLE": "HashCode"}}

Open 127.0.0.1_8000.state_machine
{"VARIABLE": "HashCode"}

Open 127.0.0.1_8001.state_machine
{"VARIABLE": "HashCode"}

Open 127.0.0.1_8002.state_machine
{"VARIABLE": "HashCode"}
```

Screenshot 4b